WEAPON TRAINING

3

CHAPTER WISE INDEX - WT (SD/SW)

Ser No	Subject	Page
1.	Chapter I : Introduction of Point 22 Rifle	127
2.	Chapter II : Handling of Point 22 Rifle	135
3.	Chapter III : Firing Operation of Rifle	149
4.	Chapter IV : Theory of Grouping and Shot Group Analysis	160
5.	Chapter V : Range Procedure and Security of Range	169
6.	Chapter VI : Short Range Firing	181

WEAPON TRAINING (SD/SW)

CHAPTER WT I : INTRODUCTION OF POINT 22 RIFLE

"Amateurs train until they get it right, professionals train until they can't get it wrong"



TEACHING INSTRUCTIONS

Period	:	One (01)
Type	:	Lecture cum Practice
Year	:	1st Year
Conducting Officer	:	Permanent Instructor.

Training Aids : Class Room, Open Training area or Ground, Script or Book Flagged or Lesson Plan in File, Board and Markers, .22 Rifles all types (01 each).

Time Plan

- Introduction : 05 Min
- Specifications of .22 Rifle (All Types) : 15 Min
- .22 Ammunition & Types : 15 Min
- Conclusion : 05 Min

INTRODUCTION

1. Hunting as a means of survival has been man's basic instinct from time immemorial. Earlier men used various types of weapons to hunt, starting gradually with bows and arrows to modern day weapons. In modern world, shooting is an exciting sport and shooting at the ranges not only helps an individual to master this sport but also gain confidence in himself. The fine art of shooting, teaches a person precision, accuracy, co-ordination of body movements, patience and confidence, all of which accrue good results in public life too with a better personality. For an NCC Cadet, introduction to weapon kindles his interest towards firing as a sport and also motivates the individual to join the Armed Forces. Hence a basic understanding of weapons is essential for Cadets.

PREVIEW

The lecture will be conducted in the following parts:-

- **Part I : Introduction.**
- **Part II : Types of Ammunition.**

LEARNING OBJECTIVES

- **Basic understanding of the point 22 rifle.**
- **Specifications of the different types of point 22 rifles used in NCC for training.**
- **Visual handling of the weapon.**
- **Understanding about the ammunition and its types.**

INTERESTING FACTS

- Popular Caliber: The Point 22 Long Rifle (LR).
- Dates back to the late 19th century.
- Point 22 LR was introduced in 1887 by the Stevens Arms Company.
- Used in training due to their manageable recoil and cost effectiveness, making them ideal for learning marksmanship.
- Popular Sporting Weapon

PART I : FAMILIARISATION OF POINT 22 RIFLE

2. The Point 22 Rifle (also depicted as .22 Rifle) is a light weight and an uncomplicated weapon. Because of its simple design and virtually no recoil, this weapon is used by NCC for firing. Like any other firearm, it is important that before using the weapon, the basic information of this weapon is available to the users.

3. **Specifications of the Point 22 Rifle**.

Details	Rifle Point 22" No II MK IV BA	Rifle Point 22" Deluxe BA	Rifle Point 22" Sporting BA
Length	45 inches	43 inches	43 inches
Weight	3.93 Kg	2.78 Kg	3 kg
Magazine Capacity	10 Rounds	05 Rounds	10 Rounds
Muzzle Velocity	2700 feet per second	2700 feet per second	2700 feet per second
Grooves in the barrel	06	06	06
Effective Range	25 yards (23 meters)	25 yards (23 meters)	25 yds (23 meters)
Max Range	1700 yds at 33° angle (1550 meters)	1700 yds at 33^{0} angle (1550 meters)	219 yards (200 meters)
Caliber	Point 22"	Point 22"	Point 22"
Ammunition	Point 22"	Point 22"	Point 22"
RATE OF FIRE			
Normal	05 rounds per minute	05 rounds per minute	05 rounds per minute
Rapid	**10-15 rounds/minute**	**10-15 rounds/ minute**	**10-15 rounds/ minute**

4. **Types of .22 Rifles Used in NCC**. Presently, three types of .22 Rifles are in use in the NCC.

(a) **Rif .22 No II Mk IV BA**



(b) **Rif .22 mm Deluxe BA**



(c) **Rif .22 Sporting BA**



5. **Parts of a .22 Rifle**. Parts of a .22 Rifle are shown below.



DESCRIPTION OF PARTS : POINT 22 RIFLE

1. **Sight.** Device used for aiming usually by aligning a front and rear sight.
2. **Muzzle**. The end of the barrel through which the projectile(bullet or shot) exits.
3. **Barrel**. Metal tube through which the projectile travels.
4. **Forestock**. Front portion of the stock extending under the barrel in front of the receiver, usually held by the non-trigger hand to help support the firearm.

5. **Magazine**. Container on a repeating firearm that holds ammunition before it's loaded into the chamber, usually tubes or boxes attached to the receiver.

6. **Trigger**. Small lever that is squeezed to start the firing process.

7. **Trigger Guard**. Piece that surrounds the trigger to protect it from being squeezed or bumped accidentally.

8. **Butt**. The part of the stock you hold against your shoulder.

9. **Stock**. Handle of Firearm.

10. **Safety**. Mechanical device that blocks the trigger or hammer to prevent accidental firing.

11. **Bolt Handle**. Handle used to open a bolt action.

12. **Bolt**. Moveable metal block that seals a cartridge into the chamber on some actions.

13. **Chamber**. Base of the barrel used to hold the cartridge or shoft shell ready for shooting.

PART II : TYPES OF AMMUNITION

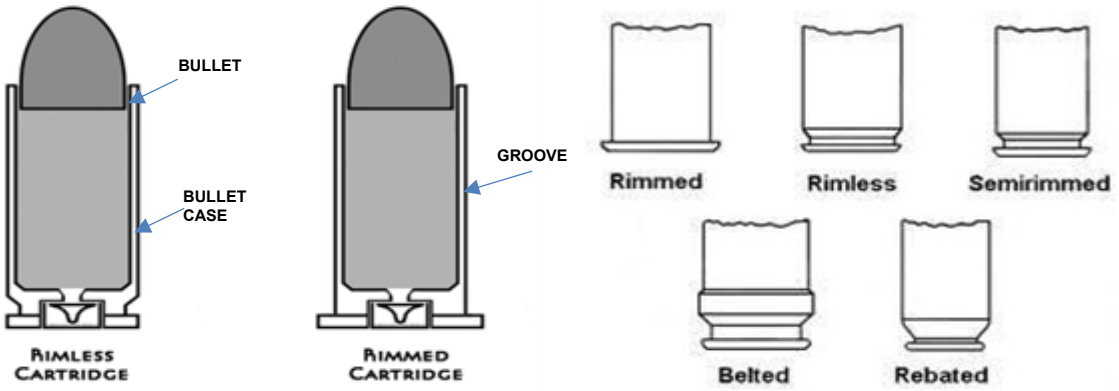
6. **Point 22 Ammunition.**

(a) The term .22 refers to the diameter of the bullet, which is approximately 0.22 inches (5.6mm). This measurement defines the caliber of the ammunition and corresponds to the size of the barrel through which it is fired.

(b) Ammunition design has evolved over time as weapon designs have become more refined. Starting with the gunpowder in the 13th century, the first cartridge made of paper, were developed in the 1500's which was later replaced by brass. Ammunition consists of a cartridge case which has a propellant and a bullet. As the propellant burns, the bullet is pushed with extreme force towards the target (will be explained in detail in further chapters). The bullets earlier were round lead balls. By the 19th century, bullets developed for rifled barrels became more aerodynamic and accurate. The .22 ammunition was first developed in 1857 in the USA.

POINT 22 AMMUNITION SPECIFICATIONS	
Caliber	Point 22 inch
Length of Bullet	10 mm
Length of Bullet with Case	15 mm
Weight	38/40 gm
Type of Bullet	Lead / Copper
Type of Ammunition	Rim / Rimless

DIAGRAMMATIC REPRESENTATION OF RIMLESS AND RIMMED AMMUNITION



7. **Difference Between Rimless and Rimmed Ammunition.**

(a) Rimmed cartridges have a rim which helps in extraction by the extractor. It engages the rim at the base of the shell which is significantly wider than the body of the cartridge.

(b) A rimless cartridge has a groove (as shown in the pic above) just above the end of the cartridge for the extractor to help in extracting. The bottom of the cartridge has the same diameter or are a little smaller (rebated) than the main body.

CONCLUSION

8. The Point 22 rifle is a highly effective and a reliable weapon. Easy to assimilate, it is the ideal weapon to start training with.

9. Cadets can achieve remarkable firing skills with good knowledge and handling of this weapon.

10. Keen students under an able instructor will have a lot to gain from this chapter.

11. Knowledge of the weapon would lead the Cadets to learn the 'BASICS' properly and thereafter gradually with more practice turn into a Marksman.

SUMMARY

- The Rifle .22 No II Mk IV BA and .22 Deluxe BA are used in NCC training.
- Both have following characteristics:
  - Caliber .22-inch LR.
  - Muzzle Velocity- 2700 feet per second. Groove: 6 Nos.
  - Effective Range- 25 yards (23 mtrs).
  - Safety : On rear end top of the body by pushing the lever forward
- Both rifles differ in their length, weight, and magazine capacity.
- These weapons fire both .22 Rimmed and Rimless ammunition.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. Length of .22 Rifle Mk-IV BA is

(a) 45 Inch (b) 43 Inch

(c) 14 Inch (d) 25 Inch

Q2. Which is not a part of .22 Rif?

(a) Bore (b) Recoil Spring

(c) Bolt (d) Piston

Q3. Effective range of .22 Rifle is _____.

(a) 50 Yds (b) 25 Yds

(c) 11 Yds (d) 23 Yds

Q4. Muzzle velocity of 0.22 rifle is _____.

(a) 2400 feet/ second (b) 1100 feet/second

(c) 2700 feet/ second (d) 2300 feet/ second

Q5. Magazine capacity of .22 Deluxe rifle magazine is _____.

(a) 05 Rounds (b) 10 Rounds

(c) 15 Rounds (d) 20 Rounds

Q6. Weight of .22 Deluxe Rifle _____.

(a) 3.6 Kg (b) 3.69 Kg

(c) 2.78 Kg (d) 3.68 Kg

Q7. How many types of ammunition are fired using the .22 Rif?

(a) 2 (b) 4

(c) 1 (d) 6

Q8 .22 Rifle was invented by which country?

(a) Russia (b) Israel

(c) India (d) USA

Q9. Normal rate of fire of .22 Deluxe is _____.

(a) 3 rounds per minute (b) 5 rounds per minute

(c) 10 rounds per minute (d) 12 rounds per minute

Q10. A primary weapon of a soldier is _____.

(a) Rifle (b) Pistol

(c) LMG (d) Mortar

Q11. .22 Rifle Mk-IV BA and Deluxe differ in _____.

(a) Length (b) Caliber

(c) Muzzle Velocity (d) Rate of Fire

Q12. Length of Bullet in .22 ammunition is _____.

(a) 05 mm (b) 10 mm

(c) 15 mm (d) 20 mm

Q13. Length of Bullet with case in .22 ammunition is _____.

(a) 05 mm (b) 10 mm

(c) 15 mm (d) 20 mm

Q14. Weight of .22 ammunition is _____.

(a) 30/32 gms (b) 35/37 gms

(c) 38/40 gms (d) 45/47 gms

WEAPON TRAINING (JD/JW)

CHAPTER WT II : HANDLING OF POINT 22 RIFLE

“"It is not the weapon that kills, but the hand and the mind that holds it."
- Napoleon Bonaparte



TEACHING INSTRUCTIONS

Period : Three (03)

Type : Lecture cum Demo/ Practice

Year : 1st Year - 02, 2nd Year - 01

Conducting Officer : Permanent Instructor.

Training Aids : Open Training area or Ground, Lesson Plan in File, Board and Markers, .22 Rifles for Squad Post Training.

Time Plan		**Ist Year**	**IInd Year**
• Introduction	:	05 Min	05 Min
• Handling of .22 Rifle	:	25 Min	10 Min
• Striping & Assembly	:	25 Min	10 Min
• Firing Positions	:	20 Min	10 Min
• Conclusion	:	05 Min	05 Min

INTRODUCTION

1. A good firer is the one, who can quickly load the magazine into Rifle, aim and fire it effectively. This can only be possible when the firer has learnt the correct procedures and carried out extensive practice during training. It is therefore important that a Cadet with practice is capable of handling the weapon in different firing positions. He should also be aware of the various parts of a Rifle and should know how to strip and assemble it. The more a Cadet dwells on these aspects, the better he or she will be in handling, and effectively using the weapon during firing practices.



PREVIEW

The lecture will be conducted in the following parts:-

- **Part I : Handling of .22 Rifle (Loading, Unloading, Aiming & Trigger Operation).**
- **Part II : Assembling & Stripping.**

LEARNING OBJECTIVES

- **Basic handling of the .22 rifle**
- **Loading & firing**
- **Aiming & trigger operation**
- **Stripping and assembling**
- **Firing positions**

INTERESTING FACTS

Shooting with both eyes open many shooters are taught to shoot with one eye closed for better focus. However, advanced shooters often keep both eyes open, which provides better peripheral vision and situational awareness in defensive situations.

PART I : HANDLING OF A POINT 22 RIFLE

2. Handling the Rifle entails **Holding, Loading, Cocking, Aiming, Trigger Operation, Firing and Unloading** the weapon while adopting various **Firing Positions**. It also entails **Stripping** & **Assembling** the weapon for cleaning and learning purposes.

3. **Holding & Loading.** In the Point 22 Deluxe Rifle, ammunition is filled one by one in the Magazine and then the Magazine is pressed into the slot made below the Rifle. Ensure that the ammunition is clean before loading. If Magazine is not available, then after assuming the firing position, the cocking handle is pulled back, the round is manually loaded into the chamber and the cocking handle is pushed forward.



4. Loading of ammunition must be carried out only on 'Orders', and no action should be taken without proper 'Orders'. Loading must be carried out properly and in a set sequence. The drill and the sequence to load a weapon using lying position is illustrated below:-

(a) To assume lying position, take a long pace forward with the left foot and at the same time pass the Rifle into left hand, holding it at the point of balance.

(b) Place the right hand on to the ground in line with the left foot. During this movement push the Rifle forward and lower it to the ground. The left arm will now be extended to the front. The legs must be keptl apart.

(c) Keep the Rifle on the right hand and move the elbow of the left hand until the target, left elbow, right shoulder and right leg come in a line. Now keep the right elbow at the place where the elbow is to the right and slightly below the shoulder.

(d) Place the palms of both hands below the chin, close your eyes and feel the tension of the elbow. If there is tension in the elbow, then keep the elbow in the same place and move the elbow back and forth to remove the tension and mark the place of the elbow.

(e) Once the position is taken, the filled magazine is loaded or the ammunition is loaded one by one directly into the chamber after every round is fired.



5. **Holding in Lying Position**. To take hold of the Rifle, push it into the shoulder very firmly. Place the Rifle at the '**V**' position formed between the pen finger and thumb of the left hand. The hand guard should come over the right hand. All four fingers should be placed from the outside and the thumb from the inside along with the magazine. To strengthen the 'hold', place the left elbow on the ground and push it a little forward. If the Rifle is pointing upwards, move the right hand forward. If it is pointing down, move the right hand backward. If it is pointing to the left then move the body position to a little right. If it is pointing to the right then move the body position to a little left.

6. **Aiming**. Accuracy of the aim is essential for a successful shot. This is by far the most difficult operation, as the eye has certain limitations. Nevertheless, good shooting can only be obtained with consistency of aim. Normally 25 yds range is used for Deluxe .22 Rifle. The thumb rule for correct aim is as enumerated below:-

(a) Focus on the target so that a clear picture is formed on the retina of the eye and one gets the true center of the target.

(b) Hold the Rifle properly as taught, keep it upright and firm, aligned to the target.

(c) Close the left eye and focus on the foresight with the right eye.

(d) See the foresight through the back sight 'U'. The foresight should be seen clearly in the center of the U. The tip of the foresight must be aligned in the center and in level with the shoulder of the U.

(e) This line formed, should further align with the point of aim.

7. **Trigger Control**. The third essential for accurate shooting is trigger operation. It should be done without disturbing the Aim. To achieve this, perfect co-ordination between eye, brain and operation of the forefinger on the trigger is required. It is achieved by conscious control of the body and thorough practice. It entails, independent action of the index finger, muscular control of the hands, holding of breath/ breath control (before pressing the trigger), co-ordination between the right eye, brain and right-hand index finger and an element of '**holding'** which essentially means hold the stance for as long as possible.

8. **Firing a Shot**. Accurate shooting can never be achieved without concentration. From the time correct holding is achieved, more than **five seconds** should not be taken to fire a shot. Dwelling too much on the aim causes the eye and muscles to tire which may result in inaccurate shooting. When the '**range'** is given, the firer should adjust the sights and await indication of the target. The sequence of firing a shot is given below:-

(a) **Position**. On being ordered to get ready, the firer must take the firing position.

(b) **Limber-Up**. Before any firing practice, it is advisable to carry out trigger operation exercise and sequence of action for firing a shot. This is termed as '**Limber-Up'** and its aim is to assist in coordination and tuning up of muscles, eye and brain.

(c) **Bhar**. Cadet should load the ammunition, take aim and be ready to fire on orders 'Bhar'.

(d) **Breathing**. Just before taking an aim, breathing must be gently restrained. It is important to coordinate, so that when the foresight comes to the point of aim, the breath is partially exhaled.

(e) **Firing**. On being ordered to Fire, the Cadet should press the trigger. For a second or two after firing, there should be no relaxation of the 'hold' or 'movement of trigger finger' or 'head'.

(f) **Follow Through**. The 'hold' on the weapon and 'point of aim' must be maintained until the bullet has left the barrel. Better still, firer should follow through until the bullet has reached the target.

(g) **Re-loading**. Immediately after follow through, on orders, reloading should be carried out.

(h) **Re-alignment**. Having reloaded, the firer should realign his sights approximately on the target and be ready to fire the next round.

9. **Unloading**. Once firing is over, order for "***Khali kar***" will be given. The magazine should be removed. The chamber be checked by pulling the cocking handle back. After this the weapon be cocked twice and on orders the trigger should be pressed keeping the rifle towards the target. On further orders, place the right hand on the ground below the right shoulder, draw up the left hand and stand up as quickly as possible.

10. **Safety Precautions**. Safety assumes paramount importance while handling any weapon. Ensuring this entails:-

(a) Pushing forward the safety catch, raising and drawing back the bolt knob, then examining the chamber and the magazine. When satisfied that both are clear, push the bolt knob forward and down, press the trigger again keeping the rifle towards the target and apply the safety catch.

(b) Inspecting the drill cartridges and ensuring that there is no live ammunition left or stuck in the barrel.

PART II : STRIPPING & ASSEMBLY

11. **Stripping**. Before stripping, check that the number on the left side of the body corresponds with the number on the back side of the lever of the bolt. The removal is done in the following sequence:-



(a) Removal of the bolt is done by raising the back sight leaf and pushing forward the safety catch. Keep the left hand under the magazine with the middle finger on the restraining catch. Withdraw the bolt to the rear, turning the bolt head upwards and fold back the back sight.

(b) Press the magazine catch upwards and take out the magazine.

12. **Assembling**. The assembling of the Rifle is always done in the reverse order. Check the magazine and ensure that the magazine number is the same as that of the Rifle. Insert the magazine at its place and press it. Ensure that the magazine is fixed in its place. While assembling the bolt of Deluxe .22 Rifle the following points will be borne in mind:-

(a) Bolt head to be fully tight.

(b) Bolt head and guide rib as well as cocking piece and steel lug to be in one line.

(c) Number inscribed on the bolt should tally with that inscribed on the Rifle.

(d) Safety catch should be applied.

(e) Insert the bolt by holding bolt lever with right hand and Rifle with the left. Push forward the bolt until it touches in the charger guide and turn the bolt head towards right till the click sound is heard. Now push forward, and press the trigger and apply the safety catch.

(f) Set the sight by pressing the thumb spring with range increasing towards the muzzle side (adjust the thumb spring accordingly).

13. **Care and Cleaning**. The efficiency of the Rifle depends on two factors.

(a) Maintenance of the Rifle.

(b) Skill of the firer.

14. The Rifle is designed to stand up to active service conditions but performance will be considerably affected if it is subjected to unduly harsh conditions. Strip the Rifle, open the butt trap and remove the pull through and the oil bottle for cleaning. The pull through has three loops. Nearest the weight is for the guage, the center for cleaning the barrel with flannelette and the end one for oiling the barrel and for use of the armourer. Cleaning to be carried out on following occasions:-

(a) **Before Firing**. Clean the barrel to make it oil free. After every practice, use the pull through to keep the barrel clean from carbon deposit.

(b) **After Firing**. Once firing is over, use the pull through to first clean the barrel of any soot, followed by using oil to keep it rust free. Open the parts and clean them before depositing the weapon in the Kote.

(c) Apart from the above, in order to keep the weapon in a fit state to fire, weekly, monthly and quarterly cleaning should be carried out.

PART III : FIRING POSITIONS

15. **Firing Positions**. You can fire from a .22 Rifle by assuming four firing positions. Keep in mind that holding, aiming and trigger operation should be correct in all of these.

(a) **Lying/Prone Position**. The prone position is a Rifle firing position where the

shooter lies on the ground. It is considered the most accurate and stable position for shooting. The main characteristics of this position are as under:-

(i) **Stability**. The prone position offers the maximum body support and stability.

(ii) **Accuracy**. It is the most accurate position for long-distance shots.

(iii) **Ease of Learning**. It is the easiest position to master and is often the first position learned.

(iv) **Fundamentals**. It is the best position for learning the fundamentals of firing such as aiming, breath control and trigger operation.



Here are some tips for using the prone position:-

- **Body Position**. Angle your body to the left axis of the Rifle if you are right-handed. Keep as much of your body in contact with the ground as possible.

- **Support**. Use a bipod if available, or put a soft rest under your front hand or under the Rifle fore-end.

- **Muzzle**. Pay special attention to the muzzle of your Rifle, as there is greater risk that it will touch the ground.

- **Rising**. Before rising (getting up), place the Rifle on the ground, stand, then pick up the Rifle.

- **Vision**. The low angle of the prone position may limit your view of the target if there are tall grasses or bushes in-between. It is therefore important to clear the field of view before proceeding with the firing practice.

(b) **Kneeling Position.** The kneeling position is a firing position where the shooter sits on the heel of one foot and places the other foot pointing forward. **There are two types of kneeling positions:-**

(i) **High Kneeling**. Sit on the heel of the foot, with the shooting-side elbow more horizontal.

(ii) **Low Kneeling**. Sit on the side of the foot, with the shooting-side foot forward.



Here are some tips for kneeling position while firing:-

- **Elbow Placement**. For right-handed shooters, place the left elbow on the left knee. For left-handed shooters, place the right elbow on the right knee.
- **Supporting Arm**. Place the other elbow under the Rifle or against the body.
- **Use a Sling**. A sling can help create a stable position.
- **Use a Stick or Bipod**. If possible, use a stick or extended bipod to steady the front hand and Rifle.
- **Pivot Point**. Think of the belt buckle as the pivot point. To engage another target, shift the entire body.
- **Relax**. Relax in the position and open the eyes to see where the target is in relation to the foresight tip.
- **Freeze**. Think of the entire body as frozen to stay in the natural point of aim.

(c) **Standing Position**. The standing position is a Rifle shooting position that's considered the most challenging and requires a lot of practice. There are two main forms of the standing position: **the supported and free-arm positions**. In the supported position, also known as the "**arm-rested position**", one puts the elbow of the non-firing hand against one's hip. In the free-arm position, one leans forward with his/her arms out in front.



Here are some tips for the standing position:-

- **Stance**. Stand perpendicular to the target with your feet shoulder-width apart and pointing slightly away from the target.

- **Arm Position**. For right-handed shooters, the left hand holds the fore stick with the elbow pointing down, and the right hand holds the grip with the elbow pointing out. The supporting arm should be perpendicular to the firearm, with the forearm resting in the supporting palm. The supporting elbow can rest on your hip or rib cage.

- **Body Position**. Keep your body upright and your legs straight with soft knees.

- **Stability**. For more stability, you can place a fist or grip the Rifle near or under the trigger guard. You can also use a stick or sticks that are perpendicular to the Rifle to increase stability.

- **Natural Point of Aim**. It is important to find your natural point of aim, which is where your body naturally aims at while aiming down range.

(d) **Sitting Position.** The sitting position is a steady shooting position that involves supporting both arms with one's legs.



Note : This position is only for teaching purpose, however, this firing position is generally not adhered to, either for practice or for any competition.

Here are some tips for firing in the sitting position:-

- **Leg position**. One can sit with one's legs apart or crossed. If the legs are apart, one can dig one's heels into the ground.
- **Arm position**. Rest the elbows on the inside of the knees, or in front of them if flexible. Avoid letting the elbows touch the kneecaps.
- **Body position**. Lean forward to rest on the legs. Turn the body 45-60 degrees away from the target.
- **Rifle position**. Place the buttstock of the Rifle on the right shoulder, close to the neck. Hold the Rifle's pistol grip firmly in the right hand to support the trigger finger.
- **Face position**. Place the face firmly against the stock, with the dominant eye looking through the sights.
- **Sling**. Use a sling to support the Rifle. Place the sling across the back of the left hand, and then across the chest and behind the left arm.
- **Steady the Rifle**. Use a stick or extended bipod to steady the fore-end of the Rifle.

CONCLUSION

16. Having learnt the characteristics of .22 Rifle, the cadets have learnt about handling the weapon.

17. Easy to handle, it is the ideal weapon to start firing with.

18. Having learnt the process of stripping, assembling and cleaning the weapon, the Cadets will be able to handle the weapon and ensure its serviceability.

19. The positions used for firing need to be further practised by the Cadets so as to be proficient at it and be fully prepared before going to the range.

SUMMARY

- **Handling the .22 Rifle**. Emphasis is laid on safe handling practices, including treating the Rifle as loaded always and being aware of the surroundings.

- **Stripping and Assembly**. It outlines the steps for safe assembly and stripping of .22 Rifle. It also highlights the importance of keeping the rifle clean along with the periodicity.

- **Firing Positions**. Covers various firing positions, such as prone, sitting, kneeling, and standing. Each position is explained in terms of stability and aiming techniques.

- Overall, the chapter aims to provide a comprehensive understanding of safe and effective Rifle handling and maintenance including various firing Positions.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. .22 Rifle has _____ parts.

(a) 12 (b) 18

(c) 15 (d) 14

Q2. How can ammunition be loaded into the rifle?

(a) Magazine (b) Manually into the chamber

(c) Both of the above (d) None of the above

Q3. When a firer is uphill and target is downhill then which position is the best for sighting the target?

(a) Standing position (b) Lying position

(c) Kneeling position (d) Sitting position

Q4. During firing, if the Rifle suddenly stops firing then what procedure is carried out?

(a) Pressing Trigger (b) Removing Magazine

(c) Put Rifle down (d) Opening Rifle

Q5. How many types of firing position are there?

(a) 3 (b) 4

(c) 5 (d) 2

Q6. What is most important for firing?

(a) Correct aiming (b) Holding tight

(c) Correct trigger operation (d) All of these

Q 7. Which part of the .22 Rifle is stripped first?

(a) Bolt (b) Magazine

(c) Re-Coil Spring (d) Receiver cover

Q 8. Which is the most stable firing position?

(a) Standing position (b) Lying position

(c) Knelling position (d) Sitting position

Q 9. How many types of kneeling positions are there

(a) 1 (b) 2

(c) 3 (d) 4

Q10. When do we need to clean a Rifle?

(a) Before putting in Kote (b) Before firing

(c) After firing (d) All of these

Q11. Chances of muzzle touching the ground is most probable in which position?

(a) Standing position (b) Lying position

(c) Knelling position (d) Sitting position

Q12. Aim of _____ is to assist in coordination and tuning up of muscles, eye and brain.

(a) Breathing (b) Trigger Control

(c) Both of the above (d) Limber Up

Q13. Before firing, the barrel should be ____.

(a) Cleaned and oil be applied (b) Cleaned and oil be removed

(c) Not cleaned (d) None of the above

Q14. After firing, the barrel should be ____.

(a) Cleaned and oil be applied (b) Cleaned and oil be removed

(c) Not cleaned (d) None of the above

Q15. Cdt should load the weapon on orders of_____.

(a) Fire (b) Khali Kar

(c) Bhar (d) Ready

Short Answer Questions

Q1. Bring out the steps in sequence of firing a shot.

Q2. How does one carry out proper aiming?

Q3. Why is prone the most accurate position for shooting?

Q4. Explain the types of kneeling positions.

Q5. Name the types of firing positions.

Long Answer Questions

Q1. How does one ensure proper prone position?

Q2. Explain the stripping and assembling process.

Q3. Explain the tips for firing in kneeling position.

WEAPON TRAINING (SD/SW)

CHAPTER WT III : FIRING OPERATION OF RIFLE

“"In the hands of a skilled marksman, the bolt-action rifle is more than a weapon-it is an extension of patience, control, and mastery."



TEACHING INSTRUCTIONS

Period	:	One (01)
Type	:	Lecture
Year	:	1st Year
Conducting Officer	:	Permanent Instructor.

Training Aids : Classroom or Squad Post Training, Charts, CBT or Audio-visual clips on Cycle of Operation and Automatic operations.

Time Plan

Introduction	:	05 Min
Part I - Brief on Cycle of Operation	:	10 Min
Part II - Introduction to Automatic Operations	:	10 Min
Part III - Introduction to Ballistics	:	10 Min
Conclusion	:	05 Min

INTRODUCTION

1. The procedures that goes into firing a round are ‘loading’, ‘cocking’, ‘firing’, ‘extraction’ and ‘ejection’. This entire thing is called cycle of operation, which can be automatic or semi-automatic. Point 22 Rifle works on bolt action operation. This cycle - chambering, cocking, firing, extracting, and reloading is manually controlled in a bolt-action rifle, giving the shooter full authority over every shot. It is a process that demands precision and deliberation, making bolt-action rifles a popular choice for accuracy dependent activities like hunting and long-range shooting.

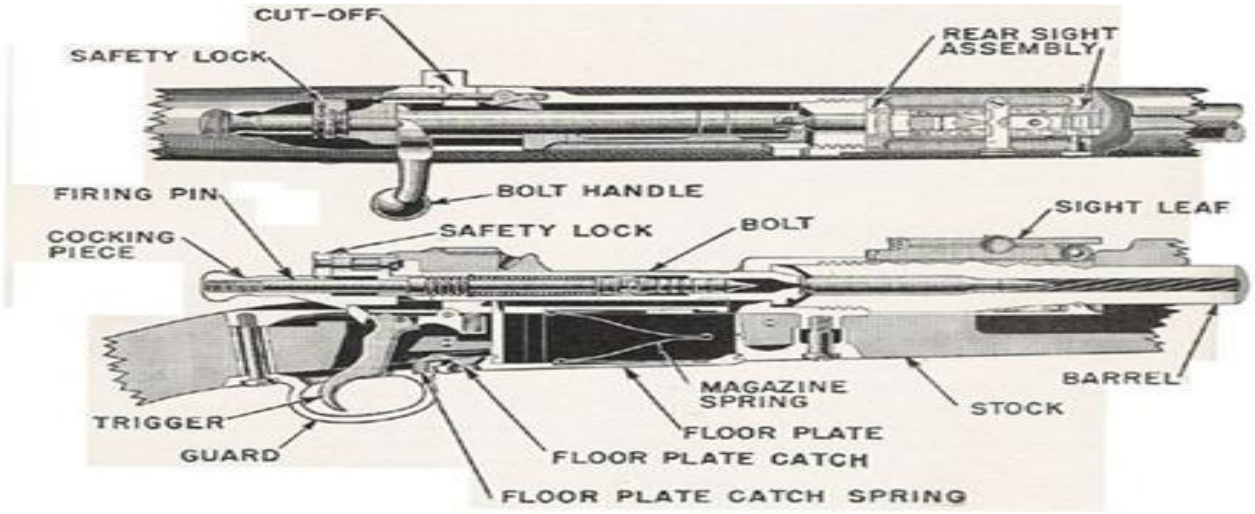
PREVIEW

The lecture will be conducted in the following parts:-

- **Part I : Brief on Cycle o operation.**
- **Part II : Introduction to Automatic Operations.**
- **Part III : Introduction to Ballistics.**

LEARNING OBJECTIVES

- **Introduction to the cycle of operation**
- **Specific to bolt action weapon**
- **Ballistics:-**
  - **Internal**
  - **External**
  - **Wound**
- **Understanding trajectory**



INTERESTING FACTS

Firing operation is generally same for all firearms, regardless of their design or action type. The steps in the cycle of operation are:-

- **Feeding** : Cartridges are inserted into the chamber.
- **Chambering** : The cartridge is moved into its final position.
- **Cocking** : The firearm locks.
- **Firing** : The firearm discharges.
- **Unlocking** : The firearm unlocks.
- **Extraction** : The cartridge is extracted.
- **Ejection** : The cartridge is ejected.

PART I : BRIEF ON CYCLE OF OPERATION

2. **Cycle of Operation**. The process of loading, cocking, firing, extracting, ejection and reloading is called cycle of operation. It is manually controlled in a bolt-action rifle. It is a process that demands precision and deliberation, making bolt-action rifles a popular choice for accuracy-dependent activities like hunting and long-range shooting.

3. In a weapon, on pressing the trigger, the bullet is fired towards the target. It moves out of the barrel with immense velocity towards the point of aim. The flight path of a bullet includes internal travel down the barrel, external path through the air, and terminal path through a target. **The science of the travel of a projectile in flight is called Ballistics**. The wounding potential of projectiles is a complex matter. The entire process can be clearly sub-divided into the following parts:-

(a) **Loading**. The process of placing the round inside the rifle either manually or in a magazine is called loading.

(b) **Cocking**. Once the cocking handle is pulled back and moved forward, the round enters the chamber and is ready to be fired.

(c) **Fire**. The moment the trigger is pressed, the hammer strikes the primer of the bullet. This gives a boost to the propellant present in the bullet case which ignites at a very high rate (Ignition of Propellant).

(d) **Production of Gas.** The controlled expansion of gases from burning propellant (gunpowder) generates pressure and the bullet is forced out of the barrel. As the bullet traverses the barrel of the gun, some minor deformation occurs, called setback deformation.

(e) Bullet travelling through a gun barrel is characterized by increasing acceleration as the expanding gases push on it, decreasing the pressure in the barrel as the bullet moves forward. Up to a point of diminishing pressure, longer the barrel, the greater the acceleration of the bullet.

(f) **Extraction and Ejection**. Once the bullet is fired, on pulling the cocking handle back, it extracts the empty case from the chamber and ejects it out of the Rifle.

PART II : INTRODUCTION TO AUTOMATIC OPERATIONS

4. Automatic operation in rifle refers to a mechanism that allows the firearm to automatically cycle and chamber the next round after each shot without the shooter manually carrying out the action. In fully automatic rifle, the rifle continues to fire rounds as long as the trigger is kept pressed. Given below is the breakdown of automatic operation in firearms.

5. **Types of Automatic Operation**.

(a) **Semi-Automatic**. In a semi-automatic rifle, the rifle one round per trigger pull. After each shot, the rifle automatically ejects the spent casing and chambers the next round, ready for the next pull of the trigger. The firer must pull the trigger for

every shot.

(b) **Fully Automatic**. In a fully automatic rifle, multiple rounds are fired continuously as long as the trigger is kept pressed. The rifle automatically cycles and chambers each subsequent round until the magazine is empty or the trigger is released. In an automatic rifle, the firing and cycling are powered by gases or recoil generated when the bullet is fired. Here is the general cycle of operation for automatic rifles:-

(i) **Gas Operation (for many modern rifles)**. As the bullet travels down the barrel, some of the expanding gases from the burning of the propellant are siphoned off into a gas tube or piston system. The gases push back on a bolt carrier or piston, cycling the bolt to the rear. As the bolt moves back, it extracts and ejects the empty case, feeds the next round and fires it. This operation continuous till the trigger is kept pressed.

(ii) **Blowback or Recoil Operation (in some designs)**. The recoil or gas pressure from the fired round forces the bolt to slide backward, ejecting the spent casing, feeding the next round and then firing it.

(iii) **Extraction and Ejection**. The bolt or slide continues moving backward, extracting and ejecting the spent cartridge case from the chamber.

(iv) **Chambering the Next Round**. As the bolt moves forward again (driven by a spring), it picks up a fresh round from the magazine and chambers it, making the rifle ready to fire again.

(v) **Resetting the Action**. For fully automatic rifles, the mechanism allows the firing pin to reset and fire the next round as long as the trigger is still being pulled.

6. **Mechanisms that Enable Automatic Operation**.

(a) **Gas-Operated Systems**. Many modern automatic and semi-automatic rifles use gas from the fired round to cycle the action, such as the AR-15/M16 series or AK-47. The gases are diverted into a system that drives the bolt back to eject the fired round and load the next one.

(b) **Blowback Operation**. It is a system of operation for self-loading firearms that obtain energy from the motion of the cartridge case as it is pushed to the rear by expanding gas created by the ignition of the propellant charge. It is commonly used in small-caliber rifles and submachine guns.

(c) **Recoil-Operated Systems**. These systems use the energy of recoil of the firearm itself to cycle the action, like in some shotguns and older automatic rifles.

(d) **Difference between Recoil Operated and Blow Back**. Recoil operated firearms use the energy of recoil to cycle the action as opposed to gas operated or blowback operations using the pressure of the propellant gases.

7. **Fully-Automatic vs Burst Fire**.

(a) **Fully-Automatic**. The rifle will continue firing as long as the trigger is pressed, cycling through the rounds in the magazine.

(b) **Burst Fire**. Some rifles are equipped with burst-fire modes (often 2 or 3 round bursts), where pulling the trigger fires a set number of rounds before stopping, requiring another pull of the trigger to fire more.

8. Examples of Automatic Rifles are M16/M4, AK-47.

9. In summary, automatic operation in rifles involves a mechanism where the firearm automatically performs the tasks of extracting, ejecting and chambering after each shot, allowing for rapid or sustained fire depending on the design (semi-automatic or fully automatic).

ADVANTAGES OF AUTOMATIC OPERATION

Faster Rate of Fire. The automatic cycling allows for a much faster rate of fire compared to manually operated firearms like bolt-action rifles.

Ease of Use in Combat. Semi-automatic and fully automatic rifles allow for more rapid response, especially in military and tactical situations.

DISADVANTAGES OF AUTOMATIC OPERATION

Less Control. Fully automatic fire can be difficult to control, often resulting in reduced accuracy due to the increased recoil and rise of the barrel.

Amn Consumption. Fully-automatic operation consumes ammunition very quickly, which can be a disadvantage in situations requiring conservation of rounds.

PART III : INTRODUCTION TO BALLISTICS

10. **Ballistics** is the science of projectiles and firearms, focusing on the movement and behavior of bullets and other objects in flight. As cadets, understanding the basics of ballistics is essential, as it helps explain how and why bullets behave differently after being fired from a weapon. Whether a person is using a rifle, a handgun, or another firearm, ballistics plays a crucial role in bullets hitting the target accurately. The flight-path of a bullet includes-internal travel inside the barrel, external path through the air and terminal path through a target. Thus, the three main types of ballistics one has to be familiar with are enumerated below:-

(a) **Internal Ballistics**. It is the study of the propulsion of a projectile and

refers to what happens inside the firearm from the moment the trigger is pressed until the bullet leaves the barrel. When the firing pin strikes the primer the propellant ignites, creating a small explosion that generates gas and builds pressure, propelling the bullet to move forward through the barrel. The length of the barrel, the condition of the firearm and the ammunition type can all affect how fast the bullet travels.

(b) **External Ballistics**. Once the bullet leaves the barrel it moves through the air, this is where external ballistics comes into play. Thus, external ballistics is that part of ballistics that deals with the behaviour of a projectile in flight. Many factors can influence the bullet's path, including gravity, air resistance, wind and the bullet's shape and speed. Understanding external ballistics helps predict where the bullet will land, which is crucial for aiming accurately, especially over longer distances. The external ballistics of a bullet's path can be determined by Initial Velocity, Gravity and Atmospheric Conditions which further determines the KE (Kinetic Energy) delivered at the target. The forward movement of the bullet is also affected by Drag (aerodynamic friction caused by air resistance). The rifling helps overcome the drag during flight. Rifling is a process of making spiral grooves inside the barrel of a gun which causes the bullets (when fired) to spin about its longitudinal longer axis to improve its aerodynamic stability and accuracy.

(c) **Terminal Ballistics**. It is the study of bullet behaviour once it impacts the target. This focuses on what happens when the bullet reaches its target. Terminal ballistics is all about how much energy the bullet transfers to the target, how deeply it penetrates and how the target reacts. Different materials-like wood, metal or a human target-will react differently based on the bullet's speed, shape and energy.

(d) **Wound ballistics**. The study of the effect of the projectile/ bullet on human body is called Wound Ballistics. It is concerning the wounding phenomenon and is part of Terminal Ballistics.

DID YOU KNOW?

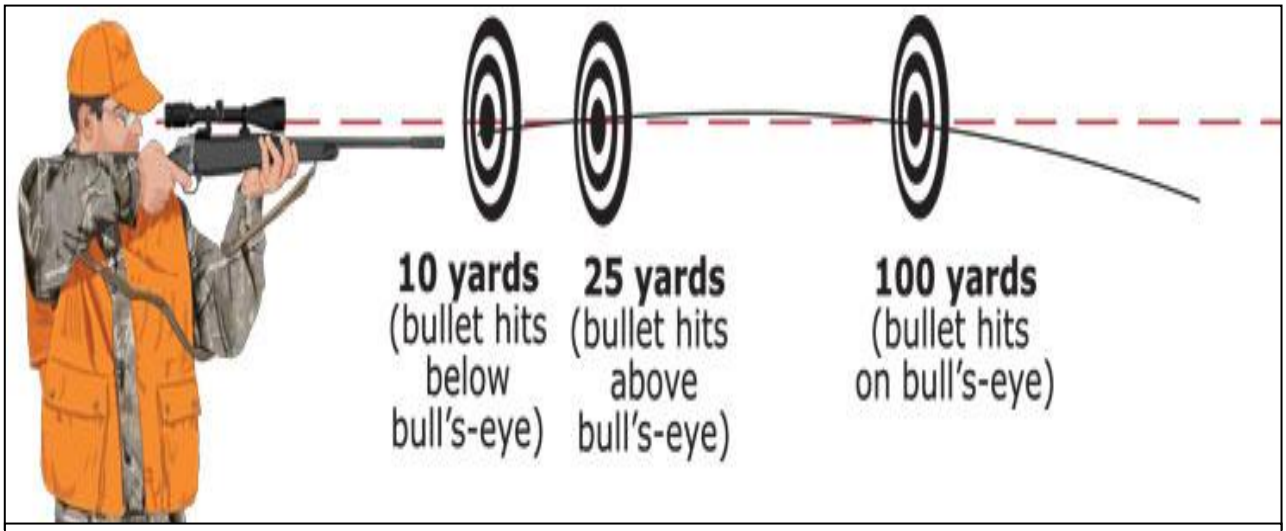
The speed at which a projectile must travel to penetrate human skin is 163 feet/second and to shatter bones is 213 feet/second, both of which are quite low, so other factors are more important in inflicting damage.

11. **Why Ballistics Matters**. A layman would always imagine a bullet travelling in a straight line. But ballistics makes us realise that it actually does not. A bullet does not travel in a straight line because of the force of gravity acting on it, causing it to drop downwards in its flight path, resulting in a curved trajectory, often described as a parabola, even from the moment it leaves the barrel; this is commonly referred to as "bullet drop". Key points are given below:-

(a) Gravity is the main factor. As soon as a bullet leaves the gun gravity pulls it downwards, causing it to deviate from a straight path.

(b) Air resistance also plays a role. While less significant than gravity, air resistance can slightly affect the bullet's trajectory by slowing it down and causing it to wobble slightly.

(c) **Impact on Aiming**. To hit a target at a distance, shooters need to aim slightly above the target to compensate for the bullet drop.



Rifle bullets don't travel in a straight line. They travel in an arc formed by the pull of gravity.

CONCLUSION

12. Each rifle operation-automatic, semi-automatic or bolt-action-serves a distinct purpose based on the needs of the shooter.

13. Whether the goal is precision firing in competitions, sustained fire in combat, quick yet controlled shots in tactical scenarios or high precision in long-range engagements, understanding the differences in operation helps in selecting the right tool for the mission.

14. The choice ultimately depends on balancing speed, accuracy and reliability according to the intended use.

15. For young cadets, mastering the fundamentals of ballistics is not just about hitting targets; it is about safety, precision and understanding the equipment you are using.

16. A good grasp of ballistics will make one an efficient marksman and help take better decisions in the field.

SUMMARY

Based on the design, weapons have the following characteristics:-

- **Automatic Rifles**. High rate of fire, effective for suppressive fire in combat, harder tocontrol due to recoil and quick ammunition depletion.

- **Semi-Automatic Rifles**. Moderate rate of fire (one shot per trigger pull), good balance of speed and accuracy, easier to control and manage recoil. Ideal for both civilian and tactical uses.

- **Bolt-Action Rifles**. Slowest rate of fire, requiring manual cycling. Highest reliability and accuracy, especially at long range. Preferred for precision shooting and hunting. Minimal mechanical complexity, leading to fewer malfunctions.

- Each type has its own strengths and is suited for different roles: automatic rifles for overwhelming firepower, semi-automatic for balanced performance and bolt-actions for precision and reliability.

- Automatic rifles typically prioritize high volume over precision, making ballistics less focused on long-range accuracy.

- Semi-automatic rifles offer a balance where understanding ballistic performance can improve shot placement over moderate distances.

- Bolt-action rifles, often used for precision shooting rely heavily on detailed ballistic knowledge to maximize accuracy over long distances.

- Ultimately, a strong grasp of ballistics is essential for maximizing performance across different rifle types, ensuring that the shooter can make effective use of the weapon's inherent capabilities.

ASSESSMENT EXERCISE

Multiple Choice Questions.

Q1. is the first step in cycle of Operation?

(a) Locking (b) Extraction
(c) Chambering (d) Firing

Q2. Science of travel of a projectile in flight is called?

(a) Firing (b) Ballistics
(c) Chambering (d) Automatic Operations

Q3. Backward movement of spent cartridge case is called?

(a) Ejection (b) Chambering
(c) Extraction (d) Locking

Q4. In Fully Automatic weapon once the trigger is pressed and held what happens?

(a) Continuous firing (b) Stop after 3 rounds
(c) Stop after a round (d) Does not fire

Q5. Study of the effect of the bullet on human body is called ---------- ballistics?

(a) Internal (b) External
(c) Terminal (d) Wound

Q6. _____ is the last step in cycle of Operations?

(a) Locking (b) Extraction
(c) Chambering (d) Ejection

Q7. In semi-automatic weapons, _____rounds are fired when trigger is pressed.

(a) 5 (b) 3
(c) 4 (d) 1

Q8. Fully automatic weapon empties the magazine when___.

(a) Trigger is pressed once (b) Trigger is pressed and left
(c) Trigger is pressed twice (d) None of the above

Q9. Blow back operation uses _____ to function.

(a) Gas (b) Force of spent cartridge
(c) Recoil of firearm (d) None of the above

Q10. Gas operation uses _____ to function.

(a) Gas (b) Force of spent cartridge
(c) Recoil of firearm (d) None of the above

Q11. Recoil operation uses _____ to function.

(a) Gas (b) Force of spent cartridge
(c) Recoil of firearm (d) None of the above

Q12. Advantage of automatic operations are.

(a) Faster rate of fire (b) Accurate firing
(c) Ammunition conservation (d) None of the above

Q13. Disadvantage of Automatic operations are.

(a) High rate of ammunition expenditure (b) Accuracy compromised
(c) Both of the above (d) Slow rate of fire

Q14. Study of effect of bullet on target is called as_____.

(a) Internal Ballistics (b) External Ballistics
(c) Terminal Ballistics (d) None of the above

Q15. Study of movement of bullet in the barrel is called as_____.

(a) Internal Ballistics (b) External Ballistics
(c) Terminal Ballistics (d) None of the above

Short Answer Questions

Q1. Name the parts of Cycle of Operations.

Q2. What are the mechanisms that enable automatic Operations?

Q3. What is the difference between fully automatic and burst fire?

Q4. List the type of ballistics.

Q5. Why does a bullet not travel in a straight line?

Long Answer Questions

Q1. What is cycle of operations? Explain the process.

Q2. Explain the cycle of operations of Fully Automatic Rifles.

Q3. What are the advantages and disadvantages of automatic operations?

WEAPON TRAINING (SD/SW)

CHAPTER WT IV : THEORY OF GROUPING & SHOT GROUP ANALYSIS



TEACHING INSTRUCTIONS

Period	:	Four (04)	
Type	:	Lecture	
Year	:	1st & 2nd Year - 02 Periods each	
Conducting Officer	:	Permanent Instructor.	
Training Aids	:	Classroom or Squad Post Training, Lesson Plan in File, Board and Markers, WT Gallery.	
Time Plan		**(1st Year)**	**(2nd Year)**
Introduction	:	05 Min	05 Min
Theory of Groups	:	30 Min	10 Min
Concept of Master Eye	:	20 Min	10 Min
Sight Alignment and Picture	:	20 Min	10 Min
Shot Group Analysis & Types of Groups	:	-	10 Min
Errors During Firing	:	-	30 Min
Conclusion	:	05 Min	05 Min

INTRODUCTION

1. The Theory of Grouping ans Shot Group Analysis is fundamental to understanding the precision and accuracy of firearms. This chapter delves into how shots are clustered (grouped) on a target and what these patterns reveal about a shooter's performance, the rifle's consistency, and external factors affecting accuracy. Grouping refers to the consistency with which multiple shots hit the target in relation to each other, rather than just how close they are to the point if aim. A small shot group indicates high precision, while a wider spread suggests variability in shooting technique or external influences. This chapter also introduces key concepts such as mean point of impact (MPI) and standard deviation, which are used to statistically analyze shot patterns. By understanding grouping and performing shot group analysis, shooters can make informed adjustments to improve their precision and overall shooting performance.

PREVIEW

The lecture will be conducted in the following parts:-

- **Part I : Theory of Groups.**
- **Part II : Concept of Master Eye.**
- **Part III : Sight Alignment & Sight Picture.**
- **Part IV : Shot Group Analysis & Types of Groups.**
- **Part V : Errors During Firing.**
- **Part VI : Conclusion.**

LEARNING OBJECTIVES

- **Theory of grouping**
- **Concepts of hat**
- **Master eye**
- **Sight alignment & sight picture**

INTERESTING FACTS

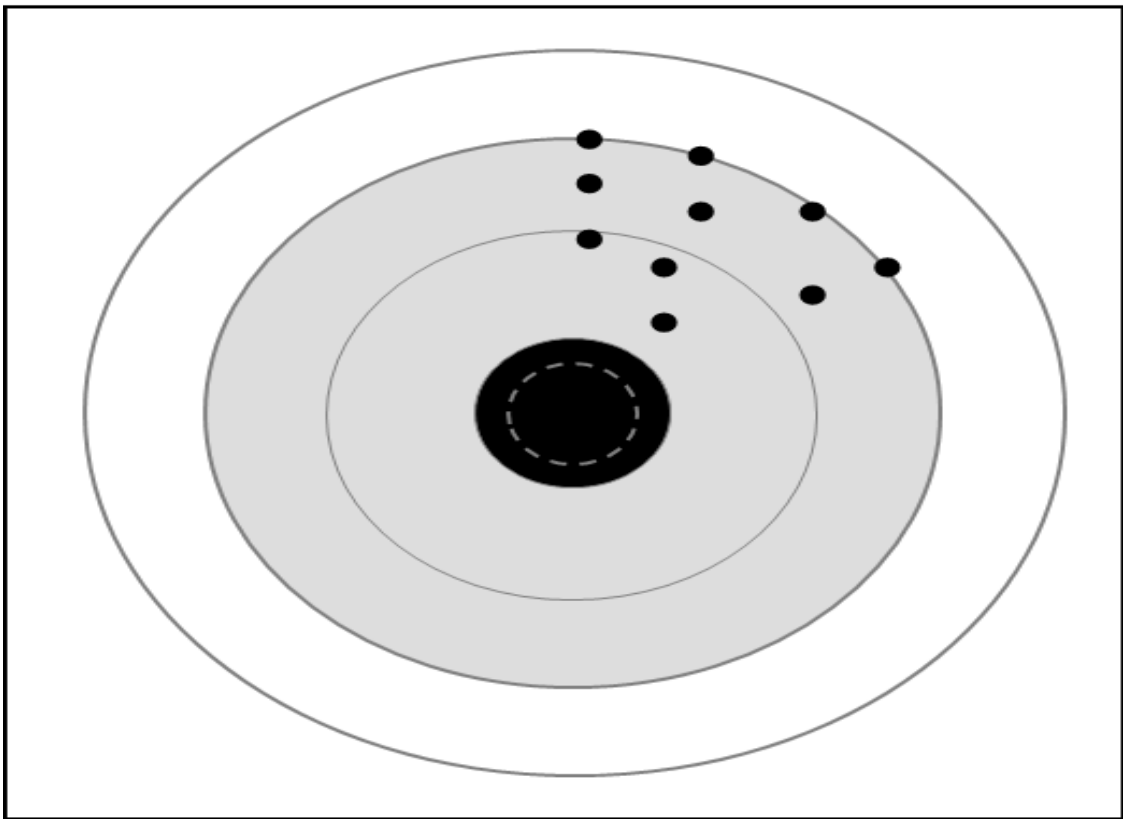
The smallest recorded shot group in history was achieved by legendary shooter Harold R. Vaughn, who achieved a five-shot group that measured only 0.009 inches at a distance of 100 yards! This level of precision is often cited as an example of what is possible with extreme attention to detail in fire arms tuning and environmental control.

In competitive shooting, sub-MOA (Minute of Angle) groups are considered the gold standard of precision. A 1 MOA group means that the shots are within a 1-inch circle at 100 yards. Elite snipers and marksmen strive to shoot even smaller groupings at much greater distances, such as 0.5 MOA or smaller.

PART I : THEORY OF GROUPS

2. **Theory of Groups.** In shooting, the theory of grouping fire is a series of shots fired at the same point of aim will form a group, rather than all hitting the exact same spot.

(a) **Grouping**. The pattern of projectile impacts on a target from multiple shots fired in one session. The width of the grouping is a measure of the weapon's precision and the shooter's skill and consistency.



(b) **Grouping Displacement**. It is the distance between the calculated center of the group and the intended point of aim. This is a measure of accuracy.

(c) **Grouping Capacity**. It is the diameter of the circle that contains all the shots fired by a person to the best of their ability.

(d) **Ammunition**. The weight and charge of the ammunition can affect the size of the group.

(e) **Number of Shots**. A group can be as few as three shots, but five shots is ideal for generating detailed feedback.

3. The size of the group is affected by:-

(a) The ammunition.

(b) The rifle.

(c) The firer.

PART II : CONCEPT OF MASTER EYE

4. **Concept of Master Eye**.

(a) Just as you have a dominant hand, you also have a dominant eye.

(b) You need to aim with the dominant-or master-eye for the most accurate shooting. Usually, your dominant eye is the same as your dominant hand, but not always. You should determine which is your dominant eye before you sight-in your rifle.

(c) To determine your dominant eye:-

(i) Form a triangular opening with your thumbs and forefingers.

(ii) Stretch your arms out in front of you.

(iii) Focus on a distant object while looking through the triangular opening and keeping both eyes open.

(iv) Bring your hands slowly to your face, keeping sight of the object through the opening; the opening will come to your dominant eye naturally.

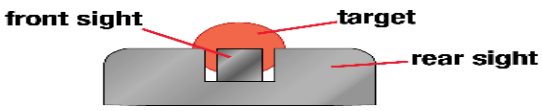
(v) If you're not sure, close one eye at a time. The weak eye will see the back of your hand; the strong one will be focused on the object in the triangle.

PART III : SIGHT ALIGNMENT AND SIGHT PICTURE

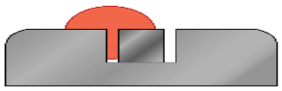
5. Sight alignment is the relationship of the front sight to the rear sight. Sight picture is the relationship of your proper sight alignment to your intended target. In other words, are your properly aligned sights aimed at the point on your target you want your bullet to hit?

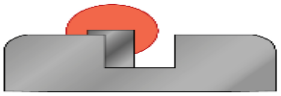
6. To obtain a proper **sight alignment**, the front sight or post is centered inside the rear sight. The top of the post should be even with the top of the rear sight.

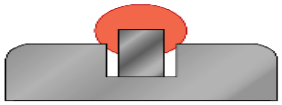
7. A proper Sight Picture is obtained when the aligned sights are put into their proper relationship with the target.

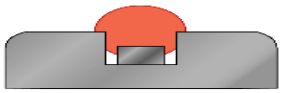


Correctly aligned: Sight is on target.


Incorrectly aligned: Bullet will go to the right of the target.


Incorrectly aligned: Bullet will go high and to the left of the target.


Incorrectly aligned: Bullet will go high.


Incorrectly aligned: Bullet will go low.

PART IV : SHOT GROUP ANALYSIS

8. **Group and its MPI**. During firing, it is important to know that when we ever fire more than one bullet, then all the bullets will not be fired at the same place on the target. There is a pattern which is formed, and there can be many reasons for it. To form a group, five bullets are fired simultaneously and at the same aiming point. The central point of the group of five bullets is called the Mean Point of Impact (MPI) of the group.

9. **Grouping Capacity**. The diameter of a circle of five bullets fired by a firer is called the grouping capacity of that firer.

(a) As per the scoring, firers are classified according to their grouping capacity, such as 2 centimeters, 4 centimeters, etc. This grouping is measured at 25 meters.

(b) It should be noted that a firer's grouping capacity is estimated correctly only if he fires each bullet with the correct shot. If a bullet is fired by a firer accidentally, without a proper screen, then that bullet should be ignored for grouping capacity.

(c) Grouping is very important for weapon zeroing. If a firer fires five bullets carrying the same point of aim (POA) and his MPI is different from the point of aim, it means zeroing of the weapon needs to be done. For zeroing, some changes are made to the foresight or backsight so that the gap between MPI and POA is reduced.

(d) Once a firing rounds are grouped, it is the responsibility of the Firer to maintain his grouping from time to time and, if possible, to improve it further.

10. **Types of Groups**. A shot group is a pattern of the effects of multiple shots fired in a single session on the target. The width of the group, or how close the shots are together, is a measure of the shooter's skill and weapon accuracy. The distance between point of aim and group calculated center point is called Grouping Displacement, which is a measure of accuracy.

(a) **Compact Group**. A Compact Group is formed when Holding, Aiming and Trigger Operations are followed correctly.

(b) **Scattered Group**. When Point of Aim and Grouping Displacement are away and the width of the group is stretched.

(c) **Bi-focal Group**. This happens when the Firer alternately focuses more on Point of Aim or Foresight Tip.



PART V : ERRORS DURING FIRING

11. **Errors During Firing**. Not being able to fire at point of Aim on a Firer's target or not making a compact group shows that the firer is making mistakes. Common mistakes are as follows:-

(a) **Bucking**. This is a common shooting mistake that occurs when a shooter tries to control the recoil force before firing. This is done by pushing the shoulder forward, inside the arms.



(b) **Flinching**. The firer anticipates a recoil and unconsciously stresses his muscles. In this case, the firer can bounce or blink as soon as the round is fired.

(c) **Trigger Jerk**. Here the firer pulls the trigger to shoot sharply, which can cause the Sight Picture to be wrong and the shot may miss the target.

12. **Reasons for Washout**. There are two main reasons for washout. Firstly, when the firer closes his master eye, and secondly when the sight alignment is wrong. It is because of these reasons that the bullet will completely miss the target.

CONCLUSION

13. The Theory of Grouping & Shot Analysis is a fundamental aspect of weapon training that enhances a marksman’s understanding of precision and consistency.

14. Grouping refers to the ability to place multiple shots within a compact area on a target, which is critical for assessing shooting skills.

15. The analysis of these shot groups allows shooters to identify patterns, diagnose errors, and make necessary adjustments to improve accuracy.

16. By knowing the errors being made while firing and then improving upon them, less or minimal mistakes are expected during firing competitions.

DID YOU KNOW?

Military snipers often practice extensive shot analysis at various distances to understand how their weapon behaves in different conditions, allowing them to make instant adjustments in the field.

SUMMARY

- **The chapter on Theory of Grouping & Shot Analysis** focuses on the principles and techniques crucial for improving shooting accuracy and consistency. Grouping refers to the ability of a shooter to place several rounds in a tight cluster on a target, demonstrating precision and control. The process of shot analysis involves studying the placement of these shots to identify shooting errors, such as incorrect sight alignment, poor trigger control, or external influences like wind and distance.

- By analyzing shot patterns, shooters can diagnose common issues like flinching, pulling the trigger incorrectly, or adjusting sights inaccurately. This analysis aids in improving one's overall shooting performance by helping to fine-tune skills.

- The chapter also emphasizes the importance of disciplined practice, where regular grouping and shot analysis allow shooters to progressively refine their skills, making adjustments as needed to achieve better accuracy. Ultimately, mastering these techniques is essential not only for individual skill development but also for operational success in military and tactical settings where precision is paramount.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. What is the pattern of projectile impacts on a target from multiple shots fired.

(a) Ballistics (b) Grouping

(c) Shot Gp (d) MPI

Q2. What is the relationship of the fore sight to the rear sight.

(a) Sight Picture (b) MPI

(c) Sight Alignment (d) Shot Analysis

Q3. Grouping is tested by firing minimum _____ rounds.

(a) 3 (b) 4

(c) 5 (d) 6

Q4. What is the Distance between pt of aim and group calculated center pt.

(a) Grouping Displacement (b) Gp picture

(c) Gp Alignment (d) Grouping Accuracy

Q5. If precision and accuracy are correct where will the bullet hit?

(a) Top center (b) Centre Left

(c) Bottom Centre (d) Pt of Aim

Q6 The diameter of the circle that contains all the shots fired by a person to the best of their ability.

(a) Grouping Capacity (b) Grouping picture

(c) Grouping Alignment (d) Grouping Accuracy

Q7 The size of the group is affected by_____.

(a) The ammunition (b) The rifle.

(c) The firer. (d) All the above

Q8 Relationship of the fore sight and rear sight to the target.

(a) Sight Picture (b) MPI

(c) Sight Alignment (d) Shot Analysis

Q9. The central point of the group of five bullets.

(a) Compact Group (b) Grouping

(c) Shot Gp (d) MPI

Q10. Weapon is zeroed if _____.

(a) MPI and POA are same

(b) MPI and POA are afar

(c) MPI is in centre and POA to right of target

(d) POA is in centre and MPI to right of target

Q11. A firer can form _____ types of Groups.

(a) 2 (b) 3

(c) 4 (d) 5

Q12. Group formed by incorrect trigger operation.

(a) Compact (b) Scattered

(c) Bi-Focal (d) None of the above

Q13. When POA and grouping displacement are close _____ Group is formed.

(a) Compact (b) Scattered

(c) Bi-Focal (d) None of the above

Q14. Firing error of pushing shoulder forward is called _____.

(a) Bucking (b) Flinching

(c) Trigger Jerk (d) None of the above

Q15. Firing error of blinking as soon as the round is fired is called _____.

(a) Bucking (b) Flinching

(c) Trigger Jerk (d) None of the above

Short Answer Questions

Q1. What are Grouping and Grouping Displacement?

Q2. What affects the size of a group?

Q3. What is difference between sight alignment and sight picture?

Q4. What is Group and its MPI?

Q5. How do you know that the firer is making mistakes?

Long Answer Questions

Q1. Explain concept of Master Eye. How do you determine it?

Q2. What are the types of Groups? Explain?

Q3. Explain the errors that occur during firing.

WEAPON TRAINING (SD/SW)

CHAPTER WT V : RANGE PROCEDURE & SECURITY OF RANGE



TEACHING INSTRUCTIONS

Period : Three (03)

Type : Lecture cum Practical

Year : Each Year- 01 Period

Conducting Officer : Permanent Instructor.

Training Aids : Classroom or Squad Post Training, Lesson Plan in File, Board and Markers, Targets and Target Papers.

Time Plan

• Introduction	:	05 Min	For all the three years
• Preparation Before Firing.	:	05 Min	
• Range Procedures	:	05 Min	
• Range Documentation	:	05 Min	
• Application of Fire	:	10 Min	
• Firing Practices	:	05 Min	
• Conclusion	:	05 Min	

INTRODUCTION

1. To maintain their shooting proficiency, troops practice at range with pistol, carbine and rifle. In our country, short ranges are found in almost every unit but classification ranges are few in numbers. Most NCC Units therefore find it difficult to find appropriate ranges for firing. If one is proficient in the steps required to be taken in order to prepare for firing, lot of time will be saved and max time available can be utilised for the basic firing practise. To take full advantage of the range allotment, preparations before the fire and executing a proper range drill at the range can give practice to all the cadets taking part in firing. Once this is done, firing practice can be carried out with full earnest. Not only can cadets be prepared for competitions but identification of good firers can also be carried out and nurtured thereafter.

PREVIEW

The lecture will be conducted in the following parts:

- **Part I : Preparation Before Firing.**
- **Part II : Range Procedures & Safety**
- **Part III : Range Documentation.**
- **Part IV : Application of Fire.**
- **Part V – Firing Practices.**
- **Part VI – Conclusion.**

LEARNING OBJECTIVES

- **Understanding range procedures**
- **Preparation for firing**
- **Security of range**
- **Essential documentation**
- **Application of fire**
- **Firing practices & competition**

INTERESTING FACTS

- Preparation before firing and executing a proper range drill is as important as firing.
- Application of fire involves using controlled fire in various training scenarios
- Grouping forms the basis of application practices
- Once the grouping capacity of the firer is assessed, it will be his duty not only to maintain this capacity, but to improve on it eradicating any faults he might have had in earlier practices.

PART I : PREPARATION BEFORE THE FIRING

2. To hone their skills and accomplish proficiency in their profession, troops practice at ranges with various small arms. To take full advantage of the range allotment, preparations before the fire and executing a proper range drill at the range can provide more time to all the trainees to carry out firing practice.

3. The range standing orders should be prepared as per the range so that the fire can start on time and there is no accident during the fire. The following points should be included towards preparations of the range:-

(a) A 20 feet high staff pole on the right edge of the stop butt with a 6 x 6 feet red flag on it.

(b) The soil of the stop butt is soft and that it does not have stones or other hard objects.

(c) Targets should be 20 feet inward from the outer edge.

(d) The stop butt should have a 2x2 feet round target number plate.

(e) A 4x4 feet red flag is placed on a 12 feet pole on the left side of the Markers gallery.

(f) Firing points and trenches should have sand bags filled with soft soil as per format.

(g) The area should be clean.

4. The following locations must be earmarked and designated on one side of the range.

(a) For collecting and depositing the Ammunition.

(b) For the Armorer with his tools and equipment.

(c) Nursing Assistant.

(d) A telephone line must be laid for the connection along with the radio set.

(e) Sentries, as per standing order, should be in red coat/sleeve.

PART II : RANGE PROCEDURE AND SAFETY PRECAUTIONS

5. **Use of Red Flags**. No firing can take place until all the red flags are hoisted and look-out men posted as per the range standing orders. One flag will always be on the top of the butt. Flags are a warning that the range is in use.

6. **Suspension of Firing**. If firing is suspended during the practice owing to some unforeseen event, weapons will be placed on the ground and the firer will stand clear.

7. **Inspection of Weapons**. The officer in charge of the firing point is responsible to ensure that all weapons are cleared and inspected before leaving the firing point.

8. A further inspection will be carried out at the conclusion of firing. During inspection, loading and unloading, all the rifles must point towards the target. On the command ***Nirikshan ke lie Janch Shastra'*** (Examine Arms) rifles will be held parallel to the ground and pointing in the direction of the target. No one will go to the target area until all weapons have been unloaded, inspected and permission to go has been given by the officer in charge.

9. Only firers, coaches (PI Staff), the firing point officer and his assistants are allowed on the firing point. Waiting details must be at least 10 yards from the firing point or in the shelter provided.

10. **The following is the normal procedure on the firing points:-**

(a) The party to fire is brought within about 100 yds of the firing point.

(b) The practices to be fired are explained.

(c) Firers are allocated target on which they have to fire.

(d) The PI staff, ammunition party and look out men take up their positions.

(e) The first two details only form up behind their respective targets on the firing point.

(f) On the order of '***Aage Barh***' (Detail Advance) the first detail will take position on the firing point.

(g) On the lowering of the red flag at the butt, the officer supervising the firing point may order his red flag to be taken down and give the order of 'Detail Advance'. The PI staff now checks up their positions and corrects them if needed and gives the word of command '***Limber Up***'. On this, the firers must align their rifles to the target. The officer will then give the order to load and Fire.

(h) The firing will start only after getting orders from officer-in-charge firing.

(j) On completion of fire, the firers must raise their right hand up, keeping the elbow on the ground.

(k) Officer-in-charge will give the word of command '***Khali Kar***' (before this he must ensure that all firers have finished firing). On this the firers will take their rifles onto their shoulders and move the bolt twice, press the trigger and stand up.

(l) The officer-in-charge will give the command 'Detail Report'. On this the firers would report '***Number ek rifle theek, do theek, teen theek and number char rifle theek***'. The report will be from left to right (Normally four targets are used on miniature range).

(m) The first and last firer will say '***Number ek/char rifle theek'*** and the rest will say '***Do theek, teen theek***' and so on.

(n) The details are changed by word of command and the new detail which has been waiting comes up while another detail forms up behind (Here the word 'detail' implies the number of firers who fire together at a point of time).

(o) Before the firers leave the range, they will have a further weapon inspection. Each firer will be asked whether he/she has any live ammunition. It will be ensured that they have no live ammunition.

11. **Safety Precautions**.

(a) Treat all rifles as loaded. Always treat a rifle as if it has a bullet in the chamber, even when it is unloaded.

(b) Always point the rifle in a safe direction, such as down range or towards the ground. The rifle is never pointed towards any individual. This is especially important when loading or unloading the rifle.

(c) Keep the finger off the trigger until ready to shoot.

(d) Know the target. Be sure of the target and what is behind, in front of it and around it.

(e) Wear eye and ear protection as appropriate.

(f) Be aware of range status. Check with others to ensure firearms are unloaded and the range is declared "SAFE" before firing. Stop shooting immediately when anyone calls "Cease fire".

(g) If there is weapon malfunction or accidental discharge, same should be reported immediately.

(h) In case of medical emergency, the Nursing Assistant must give immediate first aid and if required, the patient should be rushed to nearest hospital.

PART III : RANGE DOCUMENTATION

12. **Range Documentation**. The following documents are to be maintained at the range during firing practices:-

(a) Firing Point Register

(b) Butt Register

(c) Range Course SAO12/S/85 (new RANGE course)

(d) No Damage Certificate

(e) Ammunition and Fired Case Register

(f) Range Clearance Certificate

(g) Range Allotment Letter with timings of firing clearly specified.

(h) Before and After Firing Inspection Register

(j) Firing Record Register

PART IV : APPLICATION OF FIRE

13. In the context of range firing, application of fire involves using controlled fire in various training scenarios like target practice, Fire Control Techniques, Live-Fire Exercises, safe handling and operation of firearms including how to manage misfires or other dangerous situations, Ballistic Testing and Tactical Drills.

14. Although the Open Sight Range has the variables of 100 meters, 75 meters, 50 meters and 25 meters, but Firing in NCC is always restricted to 25 meters other than the competitions like IDSSC where the firing is done at 50 meters. However, the Zeroing will always be carried out at 25 metres.

15. Overall, the application of fire in range firing is focused on skill development, safety and tactical effectiveness. As elucidated earlier, when firing a series of shots with a supposedly constant aim and under the same conditions, all rounds will not hit the same spot however perfect the firer may be or even the weapon and its ammunition. A pattern will always be formed.

16. For the purpose of training a group will imply five consecutive well-fired shots, fired with consistent aim at the same aiming point. The central point of impact is the Mean Point of Impact (MPI) of the group.

17. **Grouping Capacity**

(a) The diameter of a circle containing all five shots, fired by the firer to the best of his ability, is known as his grouping capacity.

(b) For the purpose of coaching and scoring, firers are classified into certain grouping capacities and measured at a range of 25 meters. These capacities are represented by the diameter of the circles e.g. 2 inches, 4 inches and so on.

(c) A firer's grouping capacity, while remaining fairly constant at any given range, varies in proportion to the range at which firing is taking place. Thus, once a firer's grouping capacity at 25 meters have been ascertained, his capacity at any other range can be calculated by simple arithmetic. Hence, should the group be centrally placed, the scores expected on target can also be predicted.

18. **Declaration**. It must be appreciated that a group is representative of a firer's capacity, only when, correct aim has been taken for every shot. Should a firer accidentally fire when aim is not correct, he must note very carefully where the sights were pointing at the moment of firing and declare the fact. If on checking it is found that a bullet has gone astray and not at the place declared by the firer, it should be ignored for judging his grouping capacity. For instance, if a firer was making a 4 inches group and because of one stray bullet he makes a 12 inches group, his grouping capacity will be considered as 4 inches.

19. **Application of the Grouping**. Grouping is of immense value to the firer. It forms the basis of application practice in which the firer's Point of Aim is applied to the center of the target carefully noting the MPI, making alterations to sighting or to the point of aim where necessary. In the early stages a PI Staff will help the firer for this purpose.

20. If the position of the first shot is reasonably good and no error is declared, the second shot should be fired with the same aim. If the second shot is within the grouping capacity of the firer the probable MPI is the middle of the two shots. If MPI is too far out it should be adjusted for elevation and deflection. The correction should not be made boldly but with caution. The third shot will give the clear picture. If this MPI is again not central, a further cautious change should be made. After the fourth shot MPI position should be known and the fifth shot should be a good one fired with the correctly ascertained elevation and deflection.

PART IV : FIRING PRACTICES

21. Once the grouping capacity of the firer is assessed, it will be the duty not only to maintain this capacity, but to improve on it eradicating any faults in early shoots. The Cadets are put through the following firing practices.

22. **Miniature Range Snap Shooting.**

(a) A firer starts with grouping fire followed by application fire.

(b) After grouping and application fire has been done, the snap shooting should be carried out. This is to defeat the time factor. Initially it should be carried out on miniature range to make the firer gain confidence and realize that the exposure of target is enough to get a deliberate round fired and that the firer must be calm and composed throughout. In fact, before the fire, the firer should be given adequate practice on the following aspects: -

(i) Concentration.

(ii) Speed of co-ordination between eye and hand.

(iii) Practice in dominating impulse.

23. The practice is given in the following stages:-

(a) **Stage-1 Automatic Alignment**. The aiming mark is given at the centre when the squad is lying in semi-circle with instructor in the center. The alignment is checked with the help of aiming disc when the individual on the command `Up' aims at the Aiming Mark.

(b) **Stage-2 Automatic Alignment with Correct Hold and Trigger Operation**. On the command '***Up***' the firer brings up the rifle, aims, holds the rifle correctly and presses the trigger. He reloads when the rounds would have left the barrel for certain (It is best to emphasize this by a distinct pause), comes down and declares his shot, if incorrect, to the instructor. There is no time limit initially, only 'Accuracy before Speed' is insisted.

(c) **Stage 3, 4 & 5**. Practice firing and even competitions are carried out at figure 11 miniature target and snap shooting done at a range of 25 meters/yards.

(d) **Time**. Five exposures of seven second each is given at a regular interval over a period of 1.5 minutes. Interval between the exposures will not be less than six seconds. A trial exposure will be given to start with. Rifles may be loaded and firers may be in aimed position, before each exposure. Where facilities for pop up targets are not available, timings will be controlled from the firing point by means of a whistle. For subsequent fire, rifle will be loaded but not at aiming position. This will be raised only after the whistle is blown for each exposure. Good holding is essential.

(e) **Stage 6**. Practice is given at range, sighting at figure 11 miniature target.

(f) **State 7**. The range snap shooting practice is carried out.

CONCLUSION

24. Following range procedures and security protocols is essential for maintaining a safe environment during firing.

25. Strict adherence to the established rules not only ensures the safety of the firers and range staff but also contributes to the effectiveness of the training.

26. Proper communication, proper documentation, clear signals, and the vigilance of range officers are key factors in preventing accidents.

27. The security of the range both in terms of physical safety and the secure handling of weapons and ammunition is paramount.

28. By adhering to these guidelines, shooters can focus on their training while minimizing risks, creating a controlled and efficient learning environment.

SUMMARY

- The chapter outlines the critical protocols and safety measures necessary for ensuring safe and efficient operations during weapon training exercises on a firing range. It briefly covers standard operating procedures (SOPs) that govern the setup and conduct at a range. This includes ensuring the correct positioning of targets, establishing firing lanes, and maintaining clear communication between range officers and firers.

- A major focus is placed on safety measures, such as handling firearms responsibly, the importance of wearing protective gears (like helmet, ear and eye protection) and the significance of following all instructions from range officers. The chapter also delves into documentation including security protocols, which aims to prevent unauthorized access to the range, ensure that weapons and ammunition are accounted for, and that no live rounds are left uncollected after a session.

- Additionally, it covers emergency procedures for incidents like firearm malfunctions, accidental discharges or medical emergencies, highlighting the need for range personnel to be trained in first aid and crisis management.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. Firing is done from_____.

(a) Waiting Area (b) Hing Point
(c) Stop But (d) None of the Above

Q2. During 'Bhar' command, which action will be done at firing range.

(a) Filling bullets into magazine (b) Filling bullets into chamber
(c) Fitting magazine into Rifle (d) Trigger press

Q3. Limber up means _____.

(a) Firers must align rifles (b) Firers must start firing
(c) Firers must Stop firing (d) Firers must Check target

Q4. On orders of Fire _____action is taken: -

(a) Firer moves towards firing butt (b) Cocking the Rifle
(c) Fitting magazine in rifle (d) Starts firing

Q5. In 'Khali kar' orders, rifle will be cocked.

(a) 1 time (b) 2 times
(c) 3 times (d) 4 times

Q6. Red coloured flag at Stop Butt means_____.

(a) Start Firing (b) Stop Firing
(c) Only Grouping to be carried out (d) None of the Above

Q7. Soil on the Butt should be _____.

(a) Hard (b) Soft
(c) Mixed with stone (d) None of the above

Q8. On orders of Aage Barh

(a) Start the fire (b) Details advance
(c) Stop fire (d) Check target

Q9. On orders of “cease fire”.

(a) Firing to start (b) Firing to restart
(c) All firing to Stop (d) None of the Above

Q10. What are minimum rounds required to be fired to create a grouping?

(a) 3 (b) 4
(c) 5 (d) 6

Q11. The center point of a Group formed is called?

(a) Group (b) Grouping Capacity
(c) MPI (d) None of the above

Q12. Application fire is focused upon _____.

(a) Skill development (b) Safety
(c) Tactical effectiveness (d) All the above

Q13. Declaration is done when _____.

(a) The aim is not correct (b) The aim is correct
(c) All the above (d) None of the above

Q14. Firer should do firing in the following order:-

(a) Grouping-Application-Snap Shot (b) Snap Shot-Grouping-Application
(c) Grouping-Snap Shot- Application (d) Application-Snap Shot-rouping

Q15. Snap shot helps in defeating the _____.

(a) Time Factor (b) Handling errors
(c) Weapon malfunction (d) None of the above

Short Answer Questions

Q1. Why is preparation before firing important?

Q2. Give three pts of preparation of a range.

Q3. Why is declaring of pt of aim correctly, important?

Q4. Why is it important to fix the range correctly while firing?

Q5. What practice should be given to a firer before firing?

Long Answer Questions

Q1. What is the procedure followed for firing at the firing point?

Q2. List out the various range documents.

Q3. What are the main points to be observed in ‘Continuous Snap Shooting’?

WEAPON TRAINING (SD/SW)

CHAPTER WT VI : SHORT RANGE FIRING

"Taking my shot, one round at a time. Shooting for the stars, hitting every target. Stepping up my shooting game, one bullet at a time. When it comes to shooting, I'm a sharpshooter."





TEACHING INSTRUCTIONS

Periods	:	Nine (09).
Type	:	Practice
Year	:	2nd Year - 03 Periods; 3rd Year - 06 Periods
Conducting Officer	:	Permanent Instructor.
Training Aids	:	Short Range, Targets, Weapons, Ammunition.
Time Plan	:	0800 to 1600 hr

INTRODUCTION

1. Having undergone training of handling weapons, it's charecteristics, maintining it and the procedure and precautions to be adhered to at the ranges, the Cadets now need to undergo actual firing practice. One need to ensure that firing should be done in a phased manner so that the cadets, who are carrying out live firing for the first time, get to experience the joy of competitive firing. If trained properly, the talented kids can be identified so that better training can be provided to them.

PREVIEW

The lecture will be conducted in the following parts:

- **Part I : Grouping Fire.**
- **Part II : Application Fire.**
- **Part III : Snap Shot.**
- **Part IV : Conclusion.**

LEARNING OBJECTIVES

- **Practicing the range procedures**
- **Preparation for firing**
- **Ensure range security**
- **Firing practices & competition**

PART I : GROUPING FIRE

2. The Grouping fire will be fired as follows:-

(a) Distance : 25 Mtrs/yards

(b) Number of rounds : Five.

(c) Position : Lying with rest.

(d) Target : 1' x 1' Grouping Target.

(e) Highest Points : 40

(f) Time : Own Time

(g) **Scoring**

(i) 2.0 cms & below : 40 Points

(ii) 2.5 cms & below : 36 Points

(iii) 3.0 cms & below : 32 Points

(iv) 3.5 cms & below : 28 Points

(v) 4.0 cms & below : 24 Points

(vi) 4.5 cms & below : 20 Points

(vii) 5.0 cms & below : 16 Points

(viii) 5.5 cms & below : 12 Points

(ix) 6.0 cms & below : 08 Points

(x) 6.5 cms & below : 04 Points

(xi) Above 6.5 cms : 00 Points

PART II : APPLICATION FIRE

3. The Application fire will be fired as follows:-

(a) Distance : 25 Mtrs/ yards

(b) Number of rounds : Five.

(c) Position : Lying without rest.

(d) Target : 1' x 1' Application Target.

(e) Highest points : 40

(f) Time : Own time

(g) **Scoring**

(i) Bull : 8 Points

(ii) Inner : 6 Points

(iii) Magpie : 4 Points

(iv) Outer : 2 Points

(h) Similar practice will be carried out from Kneeling and standing (both supported and unsupported) position.

PART III : SNAP SHOOTING

4. The Snap Shot fire will be fired as follows:-

(a) Distance : 25 Meters

(b) Number of rounds : Five

(c) Position : Lying without rest

(d) Target : Figure 11 miniature

(e) Highest points : 50

(f) Time : Five exposures of seven seconds each

CONCLUSION

5. Shooting practice is important not only for sports shooting but also for self defence development.

6. In order to develop these aspirations in NCC Cdts, it is necessary that firing practice at short ranges be undertaken with utmost sincerity.

7. The short time available as also the dearth of ranges implies that we should utilize these instances to the fullest.

8. Apart from giving all the cadets an experience at firing, identification and nurturing of talent is of utmost importance.

SUMMARY

- The chapter outlines the types of practices the cadets need to undergo during firing at a range.
- Since identification of good firer is important, focus to be on grouping initially and application and snap shot subsequently.
- Additionally, the cadet gets to know the range procedures and firing precautions to be taken.